AIDOOi

AID-DATA-002 | REVISION 1

PARENT PRODUCT & VARIANT RELATIONSHIP STANDARD

Revised to align with AID-DATA-005 Product Identification & Variant Coding Standard

1. Purpose

This standard defines when multiple sellable products belong to the same underlying product family, when a new Parent Product must be created, and how Parent Products, Product/Variants and Seller Offers relate to each other. Identifier formatting follows AID-DATA-005.

2. Record Relationship

Layer	Identifier	Role
Parent Product	AIP-00000001	Logical family representing one underlying product model.
Product / Variant	AID-00000001	Canonical selectable physical version of that model.
Seller Offer	AIO-00000001	Seller-specific commercial offer linked to a Product.
Seller SKU	Seller supplied	Seller's own external inventory/reference code.

3. Core Variant Rule

Same underlying design/model + customer-selectable difference = same Parent Product.
Different fundamental model identity, structure or function = new Parent Product.

4. Variant Decision Matrix

Difference	Normally Same Parent?	Rule
Colour	Yes	Each colour receives its own AID Product ID under the same AIP Parent.
Upholstery / Fabric	Yes	Different selectable fabrics normally remain in the same family.
Finish	Yes	Different selectable finishes normally remain in the same family.
Size	Conditional	Same Parent only when the seller/manufacturer treats sizes as options of the same model.
Orientation	Usually Yes	Left-facing, right-facing or reversible versions normally remain in the same family.
Configuration	Conditional	Same Parent when configurations are official options of the same commercial model.
Fundamental design/model change	No	Create a new AIP Parent.
Major structural change	No	Create a new Parent when the underlying model identity changes.
Major functional change	Usually No	Example: standard sofa vs sofa bed normally requires separate Parent families.

5. Example - One Lounge Chair Family

A Lounge Chair model is offered in four colours. The design/model is the same; therefore all four products share one Parent.

Parent	Product ID	Variant
AIP-00000001	AID-00000001	Brown
AIP-00000001	AID-00000002	Beige
AIP-00000001	AID-00000003	Black
AIP-00000001	AID-00000004	Green

6. Example - Same Product Type, Different Family

A second Lounge Chair has a fundamentally different design/model. Its taxonomy may be identical, but it receives a new Parent Product ID.

Taxonomy	Parent	Products
Seating > Chair > Lounge Chair	AIP-00000001	AID-00000001 to AID-00000004 - Model A variants
Seating > Chair > Lounge Chair	AIP-00000005	AID-00000015 to AID-00000017 - Model B variants

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7. Product Without Variants

If a product has no genuine variants, it receives its own AID Product ID and **Parent Product ID = NULL**. Do not create an artificial Parent merely to complete the hierarchy.

Field	Example
Product ID	AID-00000018
Parent Product ID	NULL
Taxonomy	Tables > Living Room Table > Coffee Table

8. Parent Product Is Not Taxonomy

Parent Product sits below Product Type as a relationship layer. Product Type answers what the item is; Parent Product answers which underlying model family it belongs to.

Example: Seating > Chair > Lounge Chair > AIP-00000005 > AID-00000016 (Blue)

9. Parent Data vs Variant Data

Data Level	Typical Data
Parent Product	Stable model/family identity and shared model-level information.
Product / Variant	Colour, fabric, finish, variant-specific dimensions or other selectable physical differences.
Seller Offer	Seller, Seller SKU, seller price, stock, showroom/warehouse availability, delivery/commercial conditions.

10. Multi-Seller Rule

A new seller does **not** create a new Product or Parent when the physical product is the same canonical variant. A new AIO Seller Offer is created and linked to the existing AID Product.

Offer	Product	Seller	Seller SKU
AIO-00000001	AID-00000001	Blossom Home	RASO_D-6604
AIO-00000002	AID-00000001	Retailer B	CHAIR-BR-91

11. Parent Lifecycle Rules

Situation	Action
New variant added	Create new AID Product and link it to the existing AIP Parent.
Variant discontinued	Keep Product ID for history; mark relevant status inactive/discontinued.
All variants temporarily unavailable	Keep Parent and Product IDs unchanged.
Family receives new variants later	Reuse the same existing Parent.
Product is reclassified in taxonomy	Update taxonomy; Parent and Product IDs remain unchanged.
Different model introduced	Create a new Parent Product.
Identifier retired	Never reuse AIP, AID or AIO identifiers.

12. Relationship to AID-DATA-005

AID-DATA-002 defines the **business and data rules** for Parent/Variant relationships. AID-DATA-005 defines the **identifier architecture and coding format**. If an identifier-format question arises, AID-DATA-005 is the controlling standard.

13. Final Principle

AIP = Which underlying product family?
AID = Which exact selectable physical product?
AIO = Which seller is offering that product, and under what commercial conditions?

Variant relationships are determined by real product/model identity, not by similarity of ID numbers.

Revision Note

This Revision 1 replaces the earlier AID-DATA-002 coding examples based on category-embedded identifiers. The original Parent/Variant concept remains valid, while all identifiers now conform to AID-DATA-005.